

Hold-Me-Tight

Rose Ansari
Duke University
Durham, USA

2000roseansari2000@gmail.com

Abstract

HOLD-ME-TIGHT is an interactive robotic installation that examines how touch and agency are shaped by human-robot contact. Two wall-mounted robotic arms are positioned across a corner. Participants engage in two modes: a solo mode, in which a visitor uses a handheld controller to adjust the reach and grip of both arms around their own body, and a duo mode, in which one person stands between the arms while a partner operates the controller. The system emphasizes legible, bounded motion that communicates intent and maintains safety through speed limits, soft-grip trajectories, and collision-aware envelopes. Conceptually, the work proposes a model of programmed tenderness—calibrated kinematics that support careful, optionally reciprocal contact while foregrounding user consent and control. Technically, the installation uses microcontroller-driven actuators with end-effector rotation and grip mapped to a gamepad, allowing expressive yet constrained movement suitable for public exhibition. We describe design rationales, controller mappings, and evaluation criteria for comfort and perceived trust, and we outline guidelines for curating safe, comprehensible encounters with robotic embodiments in galleries and labs. HOLD-ME-TIGHT contributes to research-through-design in Human-Robot Interaction by offering a reproducible interaction grammar and safety pattern set for intimate, on-skin robotic contact.

CCS Concepts

• **Human-centered computing** → *Interaction design; Empirical studies in HCI*; • **Computer systems organization** → Robotic autonomy; • **Applied computing** → Media arts.

Keywords

interactive installation; human-robot interaction; touch; consent; safety; embodied interfaces; controller mapping; research-through-design

ACM Reference Format:

Rose Ansari. 2025. Hold-Me-Tight. In *SIGGRAPH Asia 2025 Art Gallery (SA Art Gallery '25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 1 page. <https://doi.org/10.1145/3757368.3765557>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA Art Gallery '25, Hong Kong, Hong Kong

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1883-0/25/12

<https://doi.org/10.1145/3757368.3765557>